

# Dawei Huang

Relational Generative Modeling · Adaptive Inference

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## Education and Research Experience

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### Tencent AI Lab

#### Scientific Foundation Models Group (Machine Learning Intern)

Sep 2025 – Jul 2026

Supervisor: Jianhua Yao (Tencent Distinguished Scientist)

- Developing foundation models and agent systems for protein design (**WorkBuddy MCP Server**).
- Conducting inference experiments for IgGM antibody model (**Top3** prize for AIntibody world match).

### Southern University of Science and Technology (SUSTech)

#### M.S. in Biology

Sep 2023 – Jul 2026

Advisor: Maofu Liao (Chair Professor)

Research Area: AI for Biology

### South China Agricultural University (SCAU)

#### Dual B.S. in Software Engineering and Agronomy

Sep 2019 – Jun 2023

Ding Ying Innovation Class

GPA: 4.21/5.00

Major Rank: 3rd

## Selected Research Projects

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### • Co-denoising Binder Design in a Unified Complex Latent Space

[NeurIPS 2026, Submitted]

**Core Insight:** Projecting the entire target–binder complex into a unified latent space and jointly denoising both components in a shared latent space, preserving interfacial coupling while avoiding expensive coordinate-space decoding.

#### Contributions:

- **All-atom complex representation.** Designed an interaction-aware all-atom autoencoder whose bottleneck preserves both interfacial geometry and target–binder relational topology. Achieved all-atom reconstruction DockQ greater than 0.90 on a CASP complex benchmark, substantially outperforming the baseline.
- **Latent co-denoising.** Jointly denoised the target and binder in latent space, allowing the target’s local conformation to adapt to the binder and explicitly modeling induced fit.
- **Inference-time search in latent space.** Scored and filtered intermediate latent states directly without coordinate-space decoding. Reduced inference time and GPU memory usage to 28.7% and 11.6% of Complexa, respectively, while improving designability by 20.8% over the previous version.
- **Benchmark validation.** Generated binders outperformed baseline methods in contact efficiency and local interfacial packing density, producing physically plausible geometries and competitive complex-interface quality.

### • Self-improvement Harness for Protein Binder Design

**Core Insight:** Formulate protein binder design as verifier-guided, sequential optimization of generator configurations. At each round, the controller updates input constraints, sampling budgets, and branch/rollback decisions from observed structural rewards under a fixed compute budget.

#### Contributions:

- **Strategy-level active learning.** Built a multi-agent closed-loop harness that orchestrates parameter configuration, job execution, result evaluation, and configuration updates. Introduced a composite reward and Advance/Rollback branch control to prevent linear progression along degrading search branches.
- **Fragment-template memory.** Used LLM-based structural reasoning to identify reusable local interface fragments for subsequent rounds, retaining useful motifs when global scores were unstable.
- **LLM-guided self-correction.** Introduced failure-mode attribution and feedback-driven strategy adaptation. Across eight iterations, best ipTM improved by 11% for SC2RBD and 12% for PD-L1, while the number of successful refolds increased 7-fold and 2.82-fold, respectively.

### • FANGS: Fragment-Aware Compatibility Scoring for Zero-Shot Fusion Protein Design

[Manuscript in preparation]

**Core Insight:** Fusion-protein function depends on local structural compatibility among constituent modules rather than on a global energy optimum. After controlling the functional-state conformation, compatibility

can be modeled through similarities between fragment-level latent structural embeddings, enabling efficient zero-shot ranking.

#### **Contributions:**

- **Multimodal foundation-model inference.** Designed prompt-conditioning strategies for ESM3 that enabled precise control over functional-state conformations. On fusion-protein test cases, ESM3-Template reduced structural error (RMSD) by 39% relative to AF3 predictions.
- **FANGS fragment-level scoring.** Built a compatibility score from similarities between fragment-level latent structural embeddings. Compared with global scores, FANGS increased the Spearman correlation with experimental readouts by more than 0.5.
- **Experimental antibody validation.** Affinity assays showed that the scoring trend closely tracked measured binding affinity. In nanobody design against TNF-alpha and snake-venom targets, the method identified scaffolds with higher expression yield and binding affinity.

#### **• ESM3-MD: Dynamics-Informed Conformational Sampling via Molecular-Dynamics Distributions**

[Ongoing; a related project has been submitted to NeurIPS 2026 (Third Author)]

**Core Insight:** Temporal information in molecular-dynamics trajectories can be encoded as a conditional probability distribution and injected efficiently into a pretrained model's structure module through LoRA, enabling the model to generate dynamic conformations.

#### **Contributions:**

- **Coevolution-aware fragment corpus.** Built a corpus from homologous sequence fragments to capture coevolutionary signals and optimized prompts through multi-objective combinatorial search. The optimized prompts reduced RMSD to alternative conformations by 10% and increased the diversity of sampled intermediate states by 9%.
- **ESM3-MD fine-tuning framework.** Converted temporal information from molecular-dynamics trajectories into a probability distribution and used LoRA to parameter-efficiently fine-tune the ESM3 structure module, improving conformational coverage by approximately 22% over conventional methods.

### **Publications, Manuscripts, and Patents**

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- Dawei Huang *et al.*, Submitted to NeurIPS 2026. (Protein design via latent diffusion, First Author)
- Kaiwen Cheng *et al.*, Submitted to NeurIPS 2026. (Dynamic modeling through collective supervision, Third Author)
- Jun Wu *et al.*, Submitted to NeurIPS 2026. (Cross-task screening using internal representations, Fourth Author)
- Zhu Si *et al.*, Submitted to Nature Microbiology 2026. (Flexible protein complex modeling, Fourth Author)
- Tianxu Lv *et al.*, Nat Commun. Accepted in principle. 2026. (TCR foundation model)
- Jiaqi Su *et al.*, J Cheminform., 2026. doi: 10.1186/s13321-026-01185-7 (Q1, Third Author)
- Dawei Huang *et al.*, Int. J. Mol. Sci., 2023. doi: 10.3390/ijms24108473 (Q1, First Author)
- Shiyang Zhang *et al.*, Ind Crop Prod., 2023. doi: 10.1016/j.indcrop.2023.117684 (Q1, Fourth Author)
- Patent: CN202220443530.3 (Second Inventor)

### **Technical Skills**

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- **Research:** AI for Science, Latent-Space Modeling, Inference-Time Scaling, Self-Improving Agents
- **Technical Stack:** Python, PyTorch, Autoencoders, Transformers/DiTs, Flow Matching
- **Languages:** Mandarin Chinese; English (CET-6, IELTS 6.0)

### **Honors and Awards**

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- Outstanding Poster Award, South China Structural Biology Conference
- 2026 Outstanding Graduate Student, SUSTech
- 2026 Top 10 Graduate, College of Life Sciences
- Special-Class Scholarship, SUSTech
- 2023 Outstanding Undergraduate Student, SCAU